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RECORDING

Guide to Home Recording

Everything you need to know to get professional results
from your home studio setup.

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Quality recordings begin long before mixing.

Even with the best mixing or mastering engineer, a track can only sound as good as the recordings allow. Capturing clean, balanced, and expressive performances at home is entirely possible — it just requires attention to detail and a few smart choices. This guide will help you get professional-sounding results from your home setup.

1. Room and Environment

Choose the Right Space

Pick the quietest room possible — one without noticeable echo or outside noise. Avoid rooms with large reflective surfaces (like bare walls and windows). Soft furnishings, rugs, and curtains help reduce reflections and make the space more controlled.

If possible, avoid recording near computers, air conditioners, or other noise sources.

Tip: Clap in the room — if you hear a harsh, metallic echo, it's too reflective. Add some blankets or acoustic panels.

Acoustic Treatment

You don't need a professional studio build. A few simple measures go a long way:

- Hang thick curtains or duvets behind and to the sides of your microphone.
- Place a rug underfoot if the floor is hard.
- Use a reflection filter behind the mic if available.

The goal is to minimize sound bouncing around the room before it hits the microphone.

2. Microphone Technique

Positioning and Distance

Keep about **15–20 cm (6–8 inches)** between your mouth and the microphone.

Use a **pop filter** to reduce plosive sounds (like "p" and "b").

Stay consistent in your position — don't move forward or backward mid-take.

If you're using a condenser mic, aim it slightly off-center to reduce sibilance (harsh "s" sounds).

Gain Staging

Set your recording levels so that peaks hit around **-12 dBFS to -6 dBFS** on your input meter. Avoid clipping (red lights or flat-topped waveforms). Digital clipping cannot be repaired later.

If your takes are too quiet, raise the input gain slightly — but never at the expense of distortion.

Monitoring

Use **closed-back headphones** when recording vocals or live instruments. This prevents the backing track from bleeding into your microphone.

Keep the headphone volume moderate to avoid feedback or performance fatigue.

3. Recording Instruments

Electric Guitar

Always check your tuning before every take. Guitars drift faster than you think. Even minor pitch shifts between double-tracked guitars can ruin tightness and stereo image. Fresh strings and frequent tuning make the mix and performance sound instantly more professional.

Get Your Guitar Set Up Right

Most tuning problems start before anyone hits record. Get these three things right first:

- **Match string gauge to your tuning.** Strings too light for a low or dropped tuning lose stability under hard picking, sagging in pitch right after the attack. Go a gauge or two heavier: around 11–56 for half-step-down or drop D, 12–60 for drop C or lower, and a heavy single (upper 60s to low 70s) or baritone set for drop A and below.
- **Restring right before the session, not days ahead.** An older, stretched string drops pitch noticeably as a note rings out, and it only gets worse with age. Bring new strings up to pitch without aggressively stretching them by hand; you want stable tuning for a few hours, not a “broken-in” set for a live show.
- **Use a tuner that reads in cents, and check intonation too.** A cheap clip-on tuner’s in-tune range is often wider than it looks. For intonation, play the same note open, then at the 12th fret. If the fretted note is sharp or flat against the open string, that’s a setup issue (bridge saddle adjustment), not a tuning one, and it should be checked at the tuning you’re actually recording in.

Play Tight, Then Edit Tight

No amount of mixing rescues a sloppy performance. A few habits make this much easier:

- **Rehearse until it's genuinely comfortable, before you record.** Inconsistent picking dynamics, buzzing frets, and timing slips are far easier to fix by playing the part again than by editing around them afterward.
- **Tune at the same intensity you're actually going to play with.** A string tuned while picking softly reads sharp the moment you dig in for real, especially on a low string. Picking closer to the bridge also helps. Retune often too, even if the last take sounded fine, since a guitar can drift as a whole and still sound in tune with itself.
- **Punch in on just the bad bar,** not a whole shaky pass because "the rest was good." For a genuinely difficult passage, it's fine to isolate it: loop just that phrase against a click until it's clean, then drop it in. Treat that as a last resort though, not your default. A slightly imperfect take with real character almost always beats one that's technically flawless but robotic.

Record in Sections, Not One Take

Trying to nail an entire song in a single unbroken pass almost always produces uneven energy and creeping tempo or pitch drift, especially past the two-minute mark.

- **Record in pieces.** Track verse, chorus, and bridge separately, or phrase by phrase for tricky riffs, then comp together the strongest takes. This is standard practice on professional sessions, and it's the difference between a track that sounds *played* and one that sounds *performed*.
- **Don't pile on layers for size.** Two genuinely tight, clean rhythm takes (left and right) usually sound bigger than four or five loose ones, since it's the tightness between takes that creates size, not the track count. If a part repeats identically later in the song, copy the take you already have instead of re-recording it.

Use a Real Amp or a Proper Amp Simulator

If you're tracking direct through an interface, multi-effects unit, or modeler, confirm that an actual amp simulator is engaged before you print the signal: a modeled amp and cabinet, not just compression, EQ, and time-based effects. Guitar run straight into an interface without amp/cab modeling has no speaker breakup or power-amp saturation shaping it. It reads as thin, harsh, and unmistakably "digital," no matter how good the playing is.

Before committing to a full session, record a short test pass and listen back on headphones. If it doesn't sound like a mic'd amp, your amp sim isn't actually in the signal path. Check your patch or plugin chain before you track anything for real.

Always Print a DI Track

Whether you're mic'ing a real amp or recording direct, always capture a clean **DI signal** in parallel with the processed tone.

- **It's a safety net.** This costs nothing during tracking but gives your mixer the option to re-amp or change the tone entirely without needing you back in the room.
- **It makes editing far easier.** A processed guitar track is often just a solid block of waveform, but the DI shows exactly where each note and strum starts, which matters if timing ever needs tightening. A processed-only take with no DI is a dead end if the sound needs to change later.

Label your tracks clearly inside your DAW:

- Guitar Rhythm DI
- Guitar Rhythm Amp

Give Leads Their Own Identity

A lead using the exact same guitar, tone, and settings as the rhythm tracks tends to blend in rather than cut through.

- **Change something about the tone.** Swapping guitars, or even just pickups, through the same amp setup is often enough to separate a lead from the rhythms underneath it. A touch more midrange helps it sit on top of a dense mix too.
- **Mute unused strings.** For single-note lines and octaves, lightly muting strings you're not playing with your fretting or picking hand cuts down on noise that gets buried under rhythm guitars but sticks out on an exposed lead.

Clean Up Your Edits Before You Deliver

Before a session is really done, whether you're mixing it yourself or sending it off, go back through every guitar track in solo.

- **Check every edit and crossfade** for clicks, pops, or a fade that got skipped while you were moving fast during tracking.
- **Don't snap everything to the grid.** Nudge anything that's a touch early or late, but over-tightened edits lose their feel and start to sound robotic. A little natural variation is what keeps it sounding human.
- **Trim ringing tails** so a new section hits cleanly instead of bleeding into what came before it, and keep a duplicate of your edited tracks before consolidating them down to one clean region per part.

What to Deliver to Your Mixer

A finished guitar session handed off for mixing should include:

- In-tune, edited performances comped from multiple takes, not a single raw pass with mistakes left in
- A clean DI track for every part, printed alongside any amp or amp-sim tone
- At least one reference tone print (mic'd amp or amp sim) so the mixer understands the direction you were going for

- One consolidated, edit-checked track per part, not dozens of leftover regions and crossfades from tracking
- Consistent levels with no clipping, and noise or silence trimmed between sections

If a part isn't in tune, isn't tight, or was recorded with no amp sim and no DI, no mix can fully fix it. Solving these problems is cheap during tracking. Fixing them afterward is expensive, if it's even possible at all.

Bass

Bass tuning is simpler than guitar: one string, one note at a time. But that also means there's nowhere to hide a bad note.

A few extra minutes on setup and signal chain save hours of trying to fix a bass part that's out of tune or lost in the mix.

Unlike guitar, bass doesn't need to be edited perfectly tight. A little natural feel against a locked-in grid is usually a good thing.

Track Guitars First

Tuning is the hardest thing to keep airtight on guitar. Technique, strings, and setup all fight against you, while bass is comparatively simple, since it's one note at a time.

Recording all the guitars first means that if a chord somewhere on the neck ends up a touch sharp or flat, you can nudge the bass note to match it during bass tracking, a small, easy fix. Do it the other way around and a slightly-off bass makes it nearly impossible to match a full chord shape to later. It's also easier to shape a bass tone that sits well under the guitars once they already exist, rather than guessing at how it'll fit beforehand.

Get Your Bass Set Up Right

The same setup principles as guitar apply, just scaled up.

- **Go heavier on string gauge for lower tunings.** A half-step or so down, a heaviest-available four-string set (roughly a .110 low string) is a reasonable baseline. Anything lower, go to a five-string with heavier gauges still.
- **Check intonation the same way as guitar** (open string against the 12th fret). If it reads sharp with no more room to move the saddle back, try lowering the string action instead; less finger pressure needed to fret a note means less pitch-sharpening from the press, and it often solves what the saddle alone can't.

Bass Can Sit a Little Looser

Once the guitars are locked down tight and the drums are gridded, a bass played a touch more naturally actually complements the rest of the track rather than fighting it. Bigger sections and less

note-by-note editing are fine here. Aim for solid takes and good tuning, but bass doesn't need the same tightness as rhythm guitar.

A few things worth watching for:

- **String ringing.** The low string can bleed into the next note as the player jumps between strings, a common bass habit that reads as messy against a tight mix. Mute the previous string, or split the take into smaller punches if it keeps happening.
- **High notes on a low string.** For a note fretted high up on a lower string, it's often easier to get accurate tuning by punching that note in on the next string up instead.
- **Fret-specific tuning issues.** Problems tend to show up at particular spots, like the first fret of a string. If a part lives there, check the tuning right at that fret rather than trusting a general tune-up.

Blend a Clean and a Dirty Tone

You don't need to mic a bass cab to get a great bass tone. A clean DI plus a driven or saturated tone (from a bass preamp/drive pedal or plugin) is a complete, professional-grade approach on its own.

If you're layering more than one source, each one has a job:

- **DI:** stays clean and flexible, and always gives you the option to re-amp or change the tone later without calling anyone back in.
- **Driven layer:** adds midrange aggression that helps the bass cut through dense guitars.
- **Mic'd amp** (if you have one): sits underneath as the heaviest, dirtiest of the three.

Two things to watch when blending them: check that a clean DI and a processed layer are in phase, since having traveled through different paths they can land slightly out of alignment and cancel some frequencies (listen for whichever phase relationship sounds fuller). And don't expect the dirty layer to carry the low end; it's there for midrange bite, so it's fine, often better, to roll off some of its low and extreme high end.

Label your tracks clearly inside your DAW:

- Bass DI
- Bass Drive
- Bass Amp

Clean Up Your Edits Before You Deliver

Before a session is really done, whether you're mixing it yourself or sending it off, go back through the bass track in solo and check each edit and crossfade for clicks, pops, or a missed fade.

- **Judge the edits in context,** against the rest of the rhythm section rather than in solo, so you don't over-tighten something that already sits well with the drums.
- **Reuse clean repeats.** If a section repeats and an earlier pass is noisier or looser than a later one, it's fine to copy the cleaner repeat into place rather than fixing the messier one by hand.

- **Keep a duplicate** of your edited tracks before consolidating them down to one clean region, so you can always get back to the individual takes later.

What to Deliver to Your Mixer

A finished bass session handed off for mixing should include:

- In-tune, edited performances comped from multiple takes, not a single raw pass with mistakes left in
- A clean DI track, printed alongside any driven or amp tone
- At least one reference tone print (driven or mic'd amp tone) so the mixer understands the direction you were going for
- One consolidated, edit-checked track, not dozens of leftover regions and crossfades from tracking
- Consistent levels with no clipping, and noise or silence trimmed between sections

If a part isn't in tune, isn't tight, or has no DI backing up the tone you printed, no mix can fully fix it. Solving these problems is cheap during tracking. Fixing them afterward is expensive, if it's even possible at all.

Acoustic Instruments

Use a condenser mic about 30–60 cm away, aimed where the neck meets the body for a balanced tone.

Avoid pointing the mic directly at the sound hole — this can cause boomy low-end buildup.

Record multiple takes and choose the most consistent performance, not necessarily the first.

MIDI and Virtual Instruments

If you're using software instruments, always **export the MIDI performance and a rendered audio version** once your production is complete.

This ensures the mix engineer can reproduce or tweak sounds later if needed.

Inside your DAW, keep naming consistent:

- Piano Main
- Synth Arp
- Pad Texture

4. Vocal Recording

Preparation

Warm up your voice, hydrate, and rehearse before recording.

Avoid dairy, carbonated drinks, or coffee just before vocal takes — they can affect tone and clarity.

Performance

Focus on **emotion and consistency** over perfection.

If you're recording harmonies or doubles, keep phrasing and timing tight — but let natural differences add depth.

Record multiple takes (3–5) for each section. Label them clearly in your DAW:

- Lead Vocal Take 1
- Lead Vocal Take 2
- Lead Vocal Comped (after editing)

Comping and Editing

Select the best phrases from each take and compile them into a single, strong performance.

Remove clicks, coughs, or long silences.

Avoid excessive tuning or timing correction — keep it natural unless the style demands precision.

5. File Management and Organization

Folder Structure

Keep your project tidy from the start.

A clear folder system avoids confusion when you later export for mixing.

Example:

```
/SongTitle_Session/  
├─ Audio/  
│   ├── Lead Vocal Takes/  
│   ├── Guitars/  
│   ├── Bass/  
│   ├── Drums/  
│   └─ Synths/  
├─ MIDI/  
│   ├── Piano Main.mid  
│   └─ Synth Arp.mid  
├─ RoughMixes/  
│   └─ SongTitle_RoughMix.wav  
└─ ProjectFiles/  
    └─ SongTitle.logicx
```